

K-Means

DSC 340 Applied Machine Learning
Clustering, with *k*-Means as an Example

April 4

The Plan

- Introduce unsupervised learning
- k -Means clustering algorithm
- Implementation (see PDSH online)

Supervised vs. Unsupervised Learning

Supervised

- Labeled data
- Classification methods
- Ex.: Logistic regression, *k*NN

Unsupervised

- Data may be unlabeled
- Clustering methods
- Ex.: *k*-Means (today!)
- Can be used for data exploration

k Means

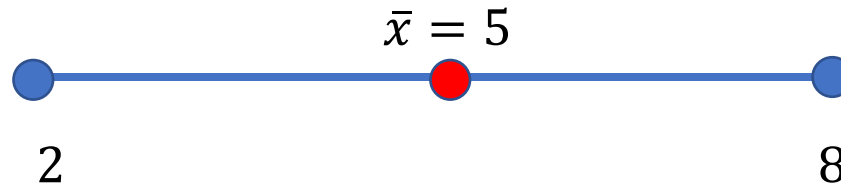
- Suppose we don't have class labels ...
- (Labeling data is often expensive!)
- But we want to cluster data into **groups of related items**
- We can choose a **distance metric** (similar to k NN)
- k -Means algorithm finds the **centroids** of **k clusters** of instances in **X**
 - Centroid: Think "midpoint" but maybe in higher dimensions
 - Average each feature over the instances (points) in the cluster

k Means Algorithm

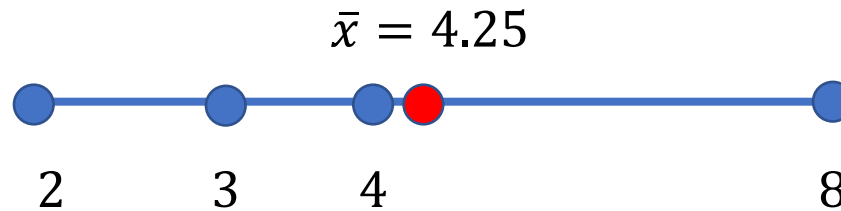
- Ahead of time: Decide on k (**hyperparameter**)
- Step 1: Choose k **random** cluster centers
 - Could choose k random points in min-max **range**
 - Or select k **data points** at random for the **seeds** (without replacement)
- Step 2: Assign each instance to the closest cluster center
 - For example, Euclidean (L2) distance
- Step 3: Recompute cluster centers
 - Compute **centroid** of instances pertaining to each cluster
- Step 4: If any cluster center moved, go to Step 2, else return clusters

Centroid in 1 dimension

- Suppose you have 2 points on a number line



- Suppose you have n points in one dimension (i.e., just one feature)



- Then centroid is just the mean (average)

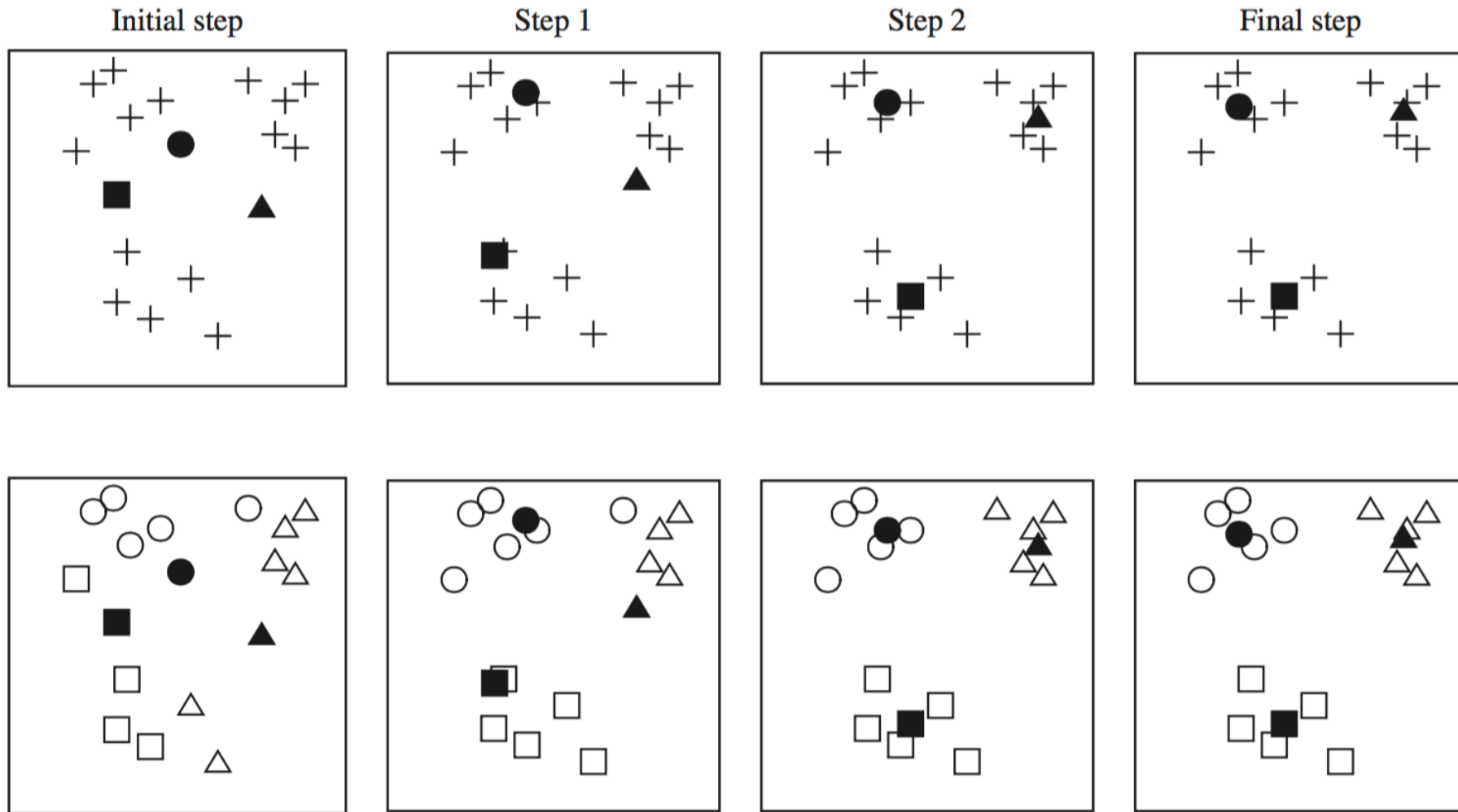
Centroid in m dimensions

- Given n training points (selected rows) ...
- with m features (columns) ...
- Find **sample mean** (average) of each column for the n points.
- The vector of means is the **centroid**.
 - Note: If we have **regions** (shapes), then we need calculus!

What the algorithm does

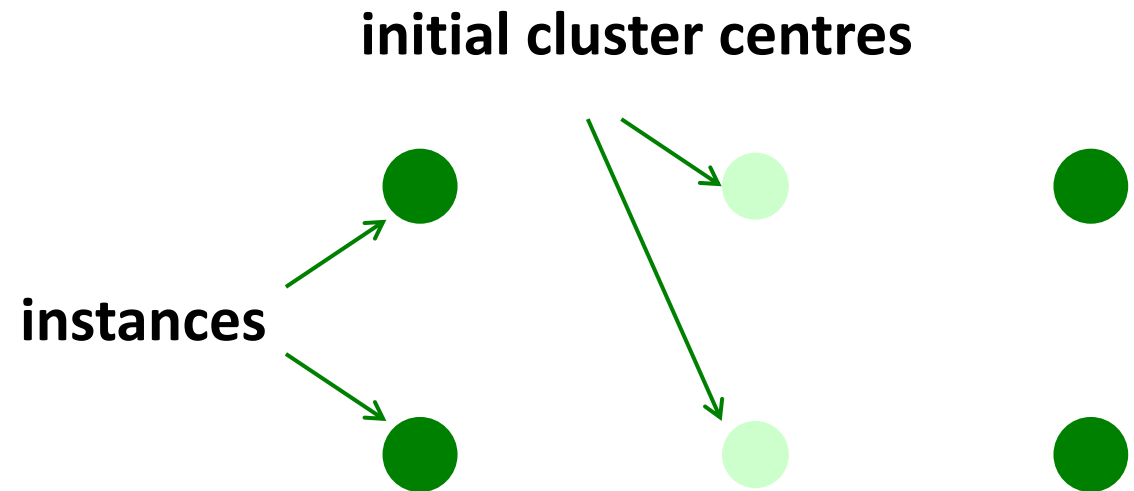
- Guaranteed to terminate
 - Converges when assignments stop changing
 - Or, equivalently, when cluster centers stop changing
 - Finite number of possible assignments
- Minimizes total squared distance ...
 - of each point from its assigned cluster center
- Converges to a **local minimum**
 -  Different initialization can yield different solutions

The k -means algorithm: example



Discussion

- Minimizes squared distance to cluster centers
- Result can vary significantly based on initial choice of seeds
- Can get trapped in local minimum
- We can use random restart to improve odds of global optimum



Implementation

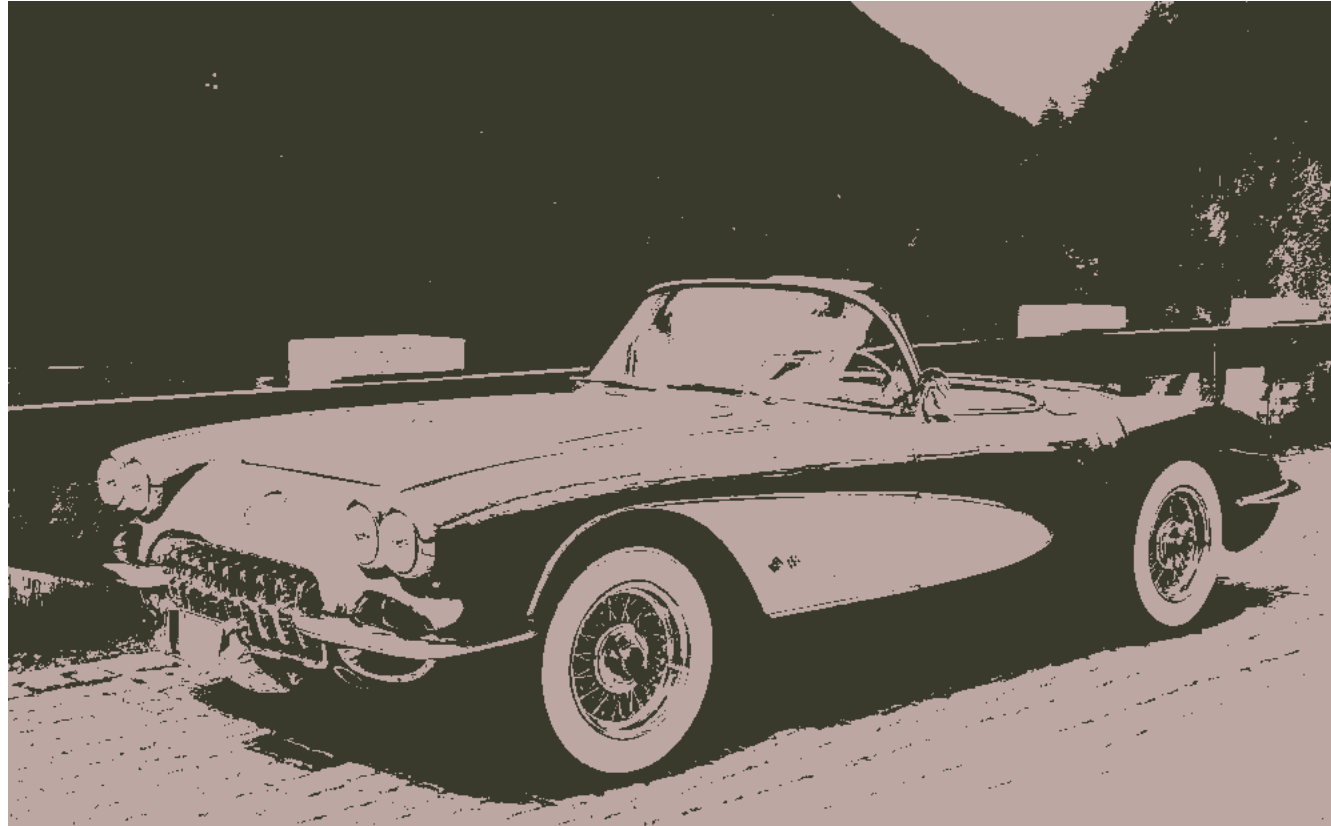
- Not hard to implement from scratch
- Alternative:
 - **from sklearn.cluster import** KMeans 😊
 - <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>

Parting thoughts

- Don't confuse kMeans and kNN!
 - Think: How are they alike? How do they differ?
- What are the parameters? What are the hyperparameters?
 - How find best value of k ?
- Related methods that achieve faster distance calculations:
 - k D-trees and ball trees
 - Hierarchical clustering, cluster into k groups then recurse

What are your questions?

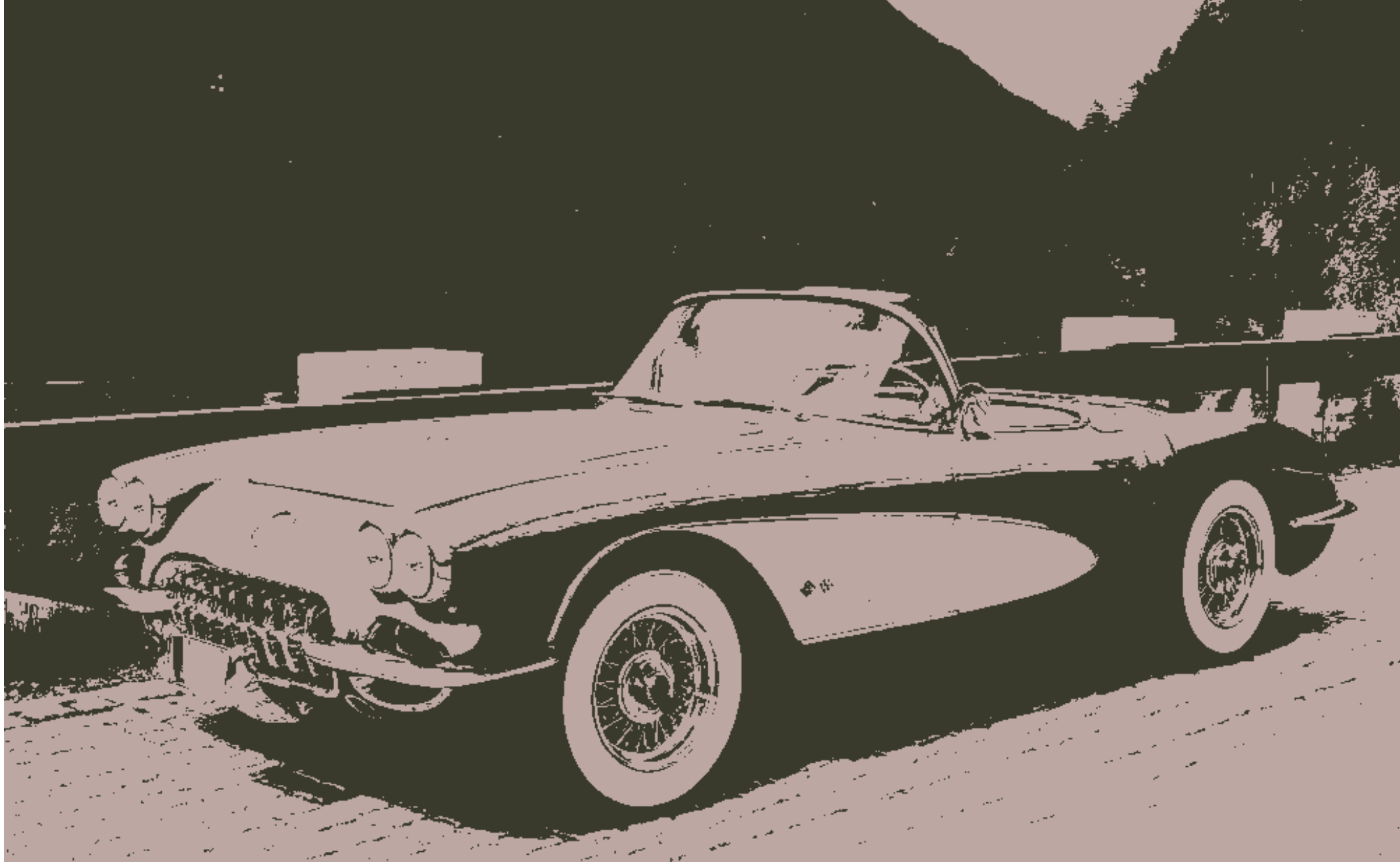
Last Lab!



Original Photo



$k=2$



k=3



k=5



$k=10$



